# Cezar-Constantin Andrici

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I'm a Ph.D. student at MPI-SP, Germany. Next, I'm going for an internship at Azure Research to work on verification of Rust programs in Lean using Separation Logic. Over the last 8 years I worked professionally in F*, Lean, Dafny and Rocq on program verification, verified compilation, and securing programs when linked with arbitrary adversarial code. My broader interests include scaling agentic proving for formal verification, protocol verification and smart contract verification. All of my papers include a full mechanization artifact, of which I am the main author. Looking for a (remote) job in Iaşi, Romania (GMT+3) starting January 2027 (negotiable).

## Education

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| 10/2021-12/2026 | **Ph.D. student** in Computer Science at **MPI-SP**, Germany<br>Subject: *Secure extraction of verified effectful F* programs to ML*. Advised by Cătălin Hriţcu. |
| 10/2019–2021 | **M.Sc.** in Computer Science from **UAIC**, Romania<br>Thesis: *Enforcing trace properties on the interoperability between F* and ML using hybrid verification*. Advised by Ştefan Ciobâcă and Cătălin Hriţcu. |
| 10/2019–2021 | **M.A.** in Regional Development from **UAIC**, Romania<br>Thesis: *Governance, Resilience and Absorption of Structural and Cohesion Funds in Central and Eastern Europe: A Case Study*. Advised by Ramona Ţigănaşu. |
| 10/2015–2019 | **B.Sc.** in Computer Science from **UAIC**, Romania<br>Thesis: *Proving SAT Solving Algorithms and Data Structures in Dafny*. Advised by Ştefan Ciobâcă. |

## Experience

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| 08/2026-10/2026<br>*(upcoming)* | **Research intern** at **Azure Research** in Security and Privacy Group, advised by Son Ho<br>Contributing to **Aeneas**, a verification tool chain for Rust programs, by extending the Lean backend with Separation Logic support and automating verification. |
| 02/2020-07/2021 | **Research intern** at **UAIC**, advised by Ştefan Ciobâcă |
| 07-10/2020 | **Research intern** at **MPI-SP** in the FOVSEC group, advised by Cătălin Hriţcu |
| 07-11/2019 | **Research intern** at **Inria Paris** in the Prosecco team, advised by C. Hriţcu and E. Rivas |
| 12/2017-06/2019 | **System Administrator** at **UAIC**, Romania<br>Responsible for maintaining the servers of the Computer Science Department, providing support to students and staff, and managing the department's IT infrastructure. Migrated the department's 15 year old main website, journal website, and research group website to WordPress. |
| 11/2015-11/2017 | **Team Lead** at **CTF365**, Cluj & Remote, Romania<br>Led the technical team for a cybersecurity startup whose VPN-based training platform let users attack target servers across a 150+ server network and served several Fortune 500 companies. Secured pre-seed investment from *hub:raum*, Deutsche Telekom's tech incubator. |
| 07-10/2015 | **Software Development Intern** at **Amazon Development Center**, Iaşi, Romania<br>Built a single-page registration application for an annual external event on top of AWS. |

## Publications

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| Conferences | **Misquoted no more: Securely extracting F* programs with IO**, C.-C. Andrici, D. Ahman, C. Hriţcu, A. Pribisova, E. Rivas, and T. Winterhalter. Conditionally Accepted, 2026 |
| | **SecRef*: Securely sharing mutable references between verified and unverified code in F***. C.-C. Andrici, D. Ahman, C. Hriţcu, R. Icleanu, G. Martínez, E. Rivas, and T. Winterhalter. *ICFP*, 2025 |
| | **Securing verified IO programs against unverified code in F***. C.-C. Andrici, Ş. Ciobâcă, C. Hriţcu, G. Martínez, E. Rivas, É. Tanter, and T. Winterhalter. *POPL*, 2024 |
| Journals | **A verified implementation of the DPLL algorithm in Dafny**. C.-C. Andrici and Ş. Ciobâcă. *Mathematics*, 2022 |

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| Informal | **Verifying non-terminating programs with IO in F$^\star$**, C.-C. Andrici, T. Winterhalter, and C. Hriţcu. HOPE, 2022 |
| | **Partial Dijkstra Monads for all**, T. Winterhalter, C.-C. Andrici, C. Hriţcu, K. Maillard, G. Martínez, and E. Rivas. TYPES, 2022 |
| | **Gradual enforcement of IO trace properties (poster)**<br>1st prize at Student Research Competition of ICFP 2020 (graduates section) |

## Research projects

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| 2023–ongoing | Lead author of **SCIO$^\star$** and **SecRef$^\star$**, two verified secure compilation frameworks for verified IO/stateful F$^\star$ programs. Proved preservation of *integrity*, *data confidentiality*, and *code confidentiality* when compiled code is linked against arbitrary adversarial unverified code. The frameworks are themselves implemented and verified in F$^\star$. |
| 2022 | **TrueSAT**: Implemented a SAT solver in Dafny and verified it to be sound, complete, and terminating. Formalized and proved correctness of the DPLL algorithm. |

## Skills

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| **Verification** | F$^\star$, Dafny, Lean, Rocq, Iris; secure compilation, program verification, SAT solving. |
| **P. Languages** | Wrote professional software in OCaml, C++, JavaScript, Ruby, SQL, PHP.<br>Other languages include Python, Haskell, Solidity and C#. |
| **Languages** | Romanian (native), English (proficient), German (beginner) |

## Teaching Assistant

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| 2024 Fall | Proofs are Programs, Graduate level course, RUB, Germany |
| 2023 Summer | Functional Programming, Undergraduate level course, RUB, Germany |
| 2020 Fall | Logics in Computer Science, Undergraduate level course, UAIC, Romania |

## Community Service

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| Sub-reviewer | CPP'26, ICFP'25, POPL'24, SP'2021 |
| AEC Member | POPL'26, POPL'23 |
| Student volunteer | POPL'24, POPL'23, ICFP'22, PLDI'20, POPL'20, ETAPS'19, FROM'18 |
| Other | Supporting the CSF 2023-2026 Test of Time Award chairs by automatically gathering and visualizing citation data for over 500 papers each year |